

Thyroid Disorders — Hypothyroidism, Hyperthyroidism and Cancer

1. Thyroid Physiology

The thyroid gland is a butterfly-shaped gland in the anterior neck that produces thyroid hormones T4 (thyroxine) and T3 (triiodothyronine). These regulate metabolism, growth, cardiac function, thermogenesis, and development of the nervous system.

The hypothalamic-pituitary-thyroid (HPT) axis regulates thyroid hormone levels. TRH (thyrotropin-releasing hormone) from hypothalamus stimulates TSH (thyroid-stimulating hormone) from pituitary → TSH stimulates T4/T3 production. T4/T3 exert negative feedback on both hypothalamus and pituitary.

T4 is the major secreted hormone and is converted to active T3 (3-5x more potent) in peripheral tissues by deiodinases. Iodine is essential for thyroid hormone synthesis — iodine deficiency is a major public health issue in India.

2. Hypothyroidism

Hypothyroidism is deficiency of thyroid hormone. Primary hypothyroidism (thyroid gland disease) is most common — elevated TSH, low free T4. Secondary is rare (pituitary failure — low TSH and low T4). Subclinical hypothyroidism: elevated TSH with normal T4.

Causes: Hashimoto's thyroiditis (autoimmune — most common in adults, positive anti-TPO antibodies), iodine deficiency, post-radioiodine/post-thyroidectomy, drugs (lithium, amiodarone, interferon), congenital.

Symptoms: fatigue, weight gain, cold intolerance, constipation, dry skin and hair, hair loss, bradycardia, hoarse voice, myalgia, muscle weakness, depression, memory impairment, menstrual irregularity, infertility, goitre.

Severe/myxoedema coma: rare life-threatening emergency — extreme hypothyroidism presenting as stupor/coma, hypothermia, hypoventilation, hyponatraemia, bradycardia, hypoglycaemia. Triggers: cold exposure, infection, sedatives. ICU management with IV levothyroxine, hydrocortisone, warming, airway management.

3. Hypothyroidism Treatment

Levothyroxine (LT4): the treatment of choice. Pure synthetic T4 taken orally once daily, 30-60 minutes before breakfast (absorption reduced by coffee, calcium, iron). Starting dose: 1.6 mcg/kg/day (full replacement) OR start low (25-50 mcg) in elderly/cardiac patients.

TSH monitoring: recheck TSH 4-6 weeks after initiating or changing dose. Target TSH: 0.5-2.5 mIU/L for most adults. Slightly higher target (1-4 mIU/L) acceptable for elderly or those with coronary artery disease.

Subclinical hypothyroidism: treat if TSH >10 mIU/L, or TSH 4.5-10 with symptoms, pregnancy, or positive anti-TPO antibodies. Some require no treatment — monitor 6-monthly.

Hypothyroidism in pregnancy: critical — foetal brain development depends on maternal thyroid hormones in first trimester. TSH targets: <2.5 mIU/L in first trimester. Screen all pregnant women. Levothyroxine dose typically needs to increase by 25-30% in pregnancy.

4. Hyperthyroidism

Hyperthyroidism: excess thyroid hormone. Thyrotoxicosis: the clinical syndrome resulting from excess thyroid hormones (any cause, including exogenous). Low TSH with elevated free T4/T3 on blood tests.

Graves' disease: autoimmune — TSH receptor-stimulating antibodies (TRAb). Most common cause of hyperthyroidism (60-80%). Classic triad: diffuse goitre, exophthalmos (proptosis — eye disease due to orbital inflammation), pretibial myxoedema (dermopathy — rare). Thyroid bruit may be present.

Toxic multinodular goitre: common in elderly in iodine-deficient areas (India, mountain regions). Toxic adenoma: single autonomous hot nodule. Thyroiditis (subacute, postpartum, drug-induced — amiodarone): release of pre-formed T4/T3; initially hyperthyroid then may become hypothyroid.

Symptoms: weight loss despite increased appetite, heat intolerance, sweating, palpitations (tachycardia, atrial fibrillation), tremor, anxiety, irritability, diarrhoea, oligomenorrhoea, muscle weakness, goitre, exophthalmos (Graves'). Apathetic hyperthyroidism in elderly: weight loss, heart failure, depression.

5. Hyperthyroidism Treatment

Anti-thyroid drugs: carbimazole (20-60 mg/day) or methimazole (15-60 mg/day) — block thyroid peroxidase enzyme. Propylthiouracil (PTU) — also blocks T4 to T3 conversion; preferred in first trimester of pregnancy. Side effects: agranulocytosis (rare but serious — check FBC if fever/sore throat), hepatotoxicity.

Beta-blockers (propranolol 40-120 mg/day or atenolol): control symptoms (palpitations, tremor, anxiety) while awaiting euthyroidism from anti-thyroid drugs. Do not treat the thyroid itself.

Radioiodine (RAI, I-131): ablates thyroid tissue. Curative in 80-90% with one dose. Standard treatment in UK/USA for Graves'. Results in permanent hypothyroidism (usually within 6-18 months). Contraindicated in pregnancy, active moderate-severe orbitopathy. Avoid close contact with children/pregnant women for 1-2 weeks post-treatment.

Thyroidectomy: indicated for large goitres causing compressive symptoms (dysphagia, stridor), suspicious nodule, Graves' with large goitre or orbitopathy, or patient preference. Total thyroidectomy preferred. Complications: hypoparathyroidism, recurrent laryngeal nerve injury.

6. Thyroid Storm (Thyrotoxic Crisis)

Thyroid storm is a rare, life-threatening exacerbation of hyperthyroidism. Precipitants: surgery, infection, trauma, contrast media, acute illness in undertreated hyperthyroid patient. Mortality 10-30% even with treatment.

Clinical features: extreme hyperthermia ($>40^{\circ}\text{C}$), tachycardia (often AF), hypertension followed by haemodynamic instability, agitation, confusion, seizures, vomiting, diarrhoea, jaundice.

Treatment (Burch-Wartofsky Point Scale guides diagnosis): PTU 500-1000 mg loading THEN 250 mg every 4 hours (blocks new synthesis AND T4→T3 conversion) OR high-dose methimazole. 1-2 hours after PTU: Lugol's iodine (SSKI) to block thyroid hormone release. IV hydrocortisone (stress doses). Propranolol IV. Treat precipitant. Active cooling. ICU admission.

7. Thyroid Nodules and Cancer

Thyroid nodules are common — palpable in 5-7% of adults, detected on ultrasound in 20-76%. The vast majority (>95%) are benign. Key concern: identify the malignant 5%.

Evaluation: thyroid ultrasound — assess nodule characteristics (size, echogenicity, margins, vascularity, calcifications, cervical lymph nodes). FNAC (fine needle aspiration cytology) — for nodules with suspicious US features (hypoechoic, irregular margins, microcalcifications, taller-than-wide shape, solid, >1 cm). Bethesda system for FNAC reporting (I-VI).

Thyroid cancers: Papillary thyroid cancer (PTC, 80-85%): most common, excellent prognosis. Follicular thyroid cancer (FTC, 10%): vascular invasion, haematogenous spread. Medullary thyroid cancer (MTC, 5%): from C-cells, produces calcitonin, associated with MEN2 syndrome. Anaplastic/undifferentiated: rare, extremely aggressive, median survival 3-5 months.

Treatment: total thyroidectomy for most cancers >1 cm. Radioiodine ablation for PTC/FTC with residual disease or metastases. TSH suppression therapy (high-dose LT4 to keep TSH <0.1) for intermediate/high-risk DTC. Targeted therapy (vandetanib, cabozantinib for MTC; sorafenib, lenvatinib for iodine-refractory DTC).

8. Iodine Deficiency in India

Iodine deficiency is the world's most prevalent, preventable cause of brain damage and intellectual disability. Iodine deficiency causes: goitre (most visible manifestation), hypothyroidism, cretinism (congenital hypothyroidism — irreversible intellectual disability if untreated), increased risk of thyroid cancer.

India had one of the world's largest iodine deficiency problems. Universal Salt Iodisation (USI) — mandatory iodization of edible salt (15-30 ppm iodine) — is the cornerstone of India's National Iodine Deficiency Disorders Control Programme (NIDDCP). India largely achieved iodine sufficiency in most states, though pockets remain in Himalayan regions and inland areas.

Goitre belt: Sub-Himalayan belt stretching from Jammu and Kashmir through Uttar Pradesh, Bihar, Jharkhand, West Bengal, and North-Eastern states. Iodine-poor soil → iodine-poor food and water. Regular monitoring of urinary iodine excretion (UIE) and goitre prevalence.

9. Thyroid in Pregnancy

Pregnancy profoundly affects thyroid physiology: increased TBG (thyroxine-binding globulin), increased T4 demand, iodine loss via kidneys. TSH is suppressed in first trimester by hCG (cross-reacts with TSH receptors).

Hypothyroidism in pregnancy: associated with miscarriage, pre-eclampsia, placental abruption, premature birth, low birth weight, impaired foetal neurodevelopment (IQ reduction). Treat promptly to maintain TSH <2.5 mIU/L in first trimester, <3 mIU/L later.

Hyperthyroidism in pregnancy: Graves' disease. PTU preferred in first trimester (methimazole associated with aplasia cutis, choanal atresia). Switch to methimazole/carbimazole in second trimester (PTU hepatotoxicity). Uncontrolled hyperthyroidism: foetal loss, preterm labour, small-for-gestational-age, neonatal thyrotoxicosis (maternal TRAb crosses placenta).

Postpartum thyroiditis: affects 5-9% of women. Autoimmune thyroid inflammation post-partum. Transient hyperthyroid phase (1-4 months) followed by hypothyroid phase (4-8 months), usually recovering by 12 months. Some develop permanent hypothyroidism.

10. Thyroid Health — Screening and Awareness

TSH testing is the single best screening test for thyroid dysfunction. Universal newborn TSH screening to detect congenital hypothyroidism (CH): TSH heel-prick on day 3-5 of life. Early LT4 treatment prevents permanent intellectual disability. India's Rashtriya Bal Swasthya Karyakram (RBSK) includes thyroid screening.

Opportunistic screening: women over 35, those with family history of autoimmune disease, patients on interferon, lithium, or amiodarone, people with unexplained anaemia, hyponatraemia, or elevated lipids.

Awareness: thyroid conditions are significantly underdiagnosed in India, particularly in women. Campaigns by the Indian Thyroid Society and public health initiatives increase awareness. Serum TSH cost has fallen significantly — widely available in most laboratories at ≤₹200.

Self-examination: patients should be taught neck self-examination. A rapidly enlarging nodule, hard texture, dysphonia (recurrent laryngeal nerve invasion), or cervical lymphadenopathy are

red flags for thyroid cancer requiring urgent evaluation.